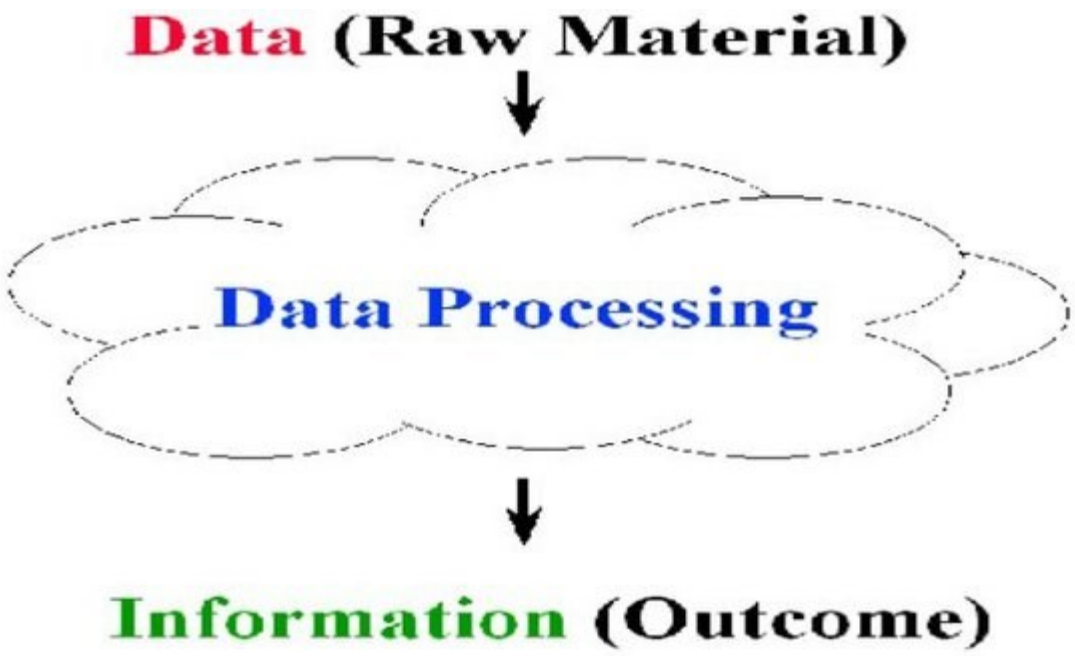
Information Systems Brief Discussions

By
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Information Concepts (1)

- Data, Information, system, information system



Characteristics of Valuable Information

- accurate,
- complete,
- cost effective (economical),
- current (timely)
- user friendly
- relevant
- Reliable
- accessible,
- easily to
- flexible,
- simple,
- verifiable,
- secure



Uses of Information

to achieve org. goal

Information support **to achieve desired objectives**

- Planning, organizing, decision making, …

- Information/systems for control
- Information /systems for planning
- Information/systems for decision making
- Information for competitive advantage
- Reduce **“Uncertainty”**

System

- Definition
  - A set of elements or components that interact to accomplish goals

  -Set of interrelated components
  - Collect, process, store, and distribute information

  →By computers and software as a tool
  - Support decision making, coordination, control, problem analysis and create new product
  - Provide solutions to challenges in business Env.

inser

System Components and Concepts

- System boundary
  - Defines the system and distinguishes it from everything else
- System types
  - Simple vs. complex
  - Open vs. closed –
    - Open system: System that interfaces with other systems
    - Closed system: Stand-alone system that has no contact with other systems)
  - Stable vs. dynamic
  - Adaptive vs. non-adaptive
  - Permanent vs. temporary

System Performance and Standards

- Efficiency
  - A measure of what is produced divided by what is consumed (e.g. Efficiency of a motor is the energy produced divided by what is consumed)
- Effectiveness
  - A measure of the extent to which a system achieves its goals
- System performance standard
  - A specific objective of the system

System Variables and Parameters

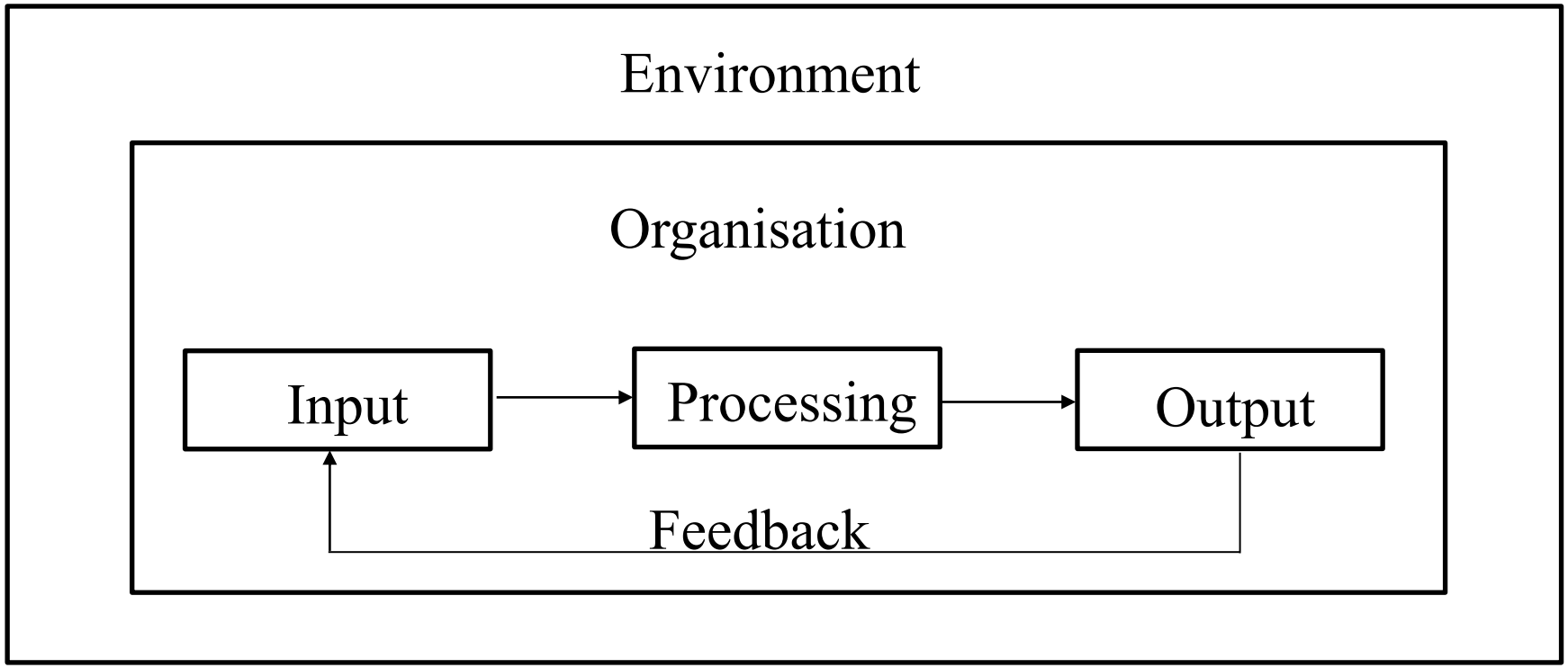
- System variable
  - A quantity or item that can be controlled by the decision maker
  - E.g. the price a company charges for a product

- System parameter
  - A value or quantity that cannot be controlled by the decision maker
  - E.g., cost of a raw material

Information System (IS)

- Definition
  - **A set of interrelated elements or components that collect (input), manipulate (process), and disseminate (output) data and information and provide a feedback mechanism to meet an objective**

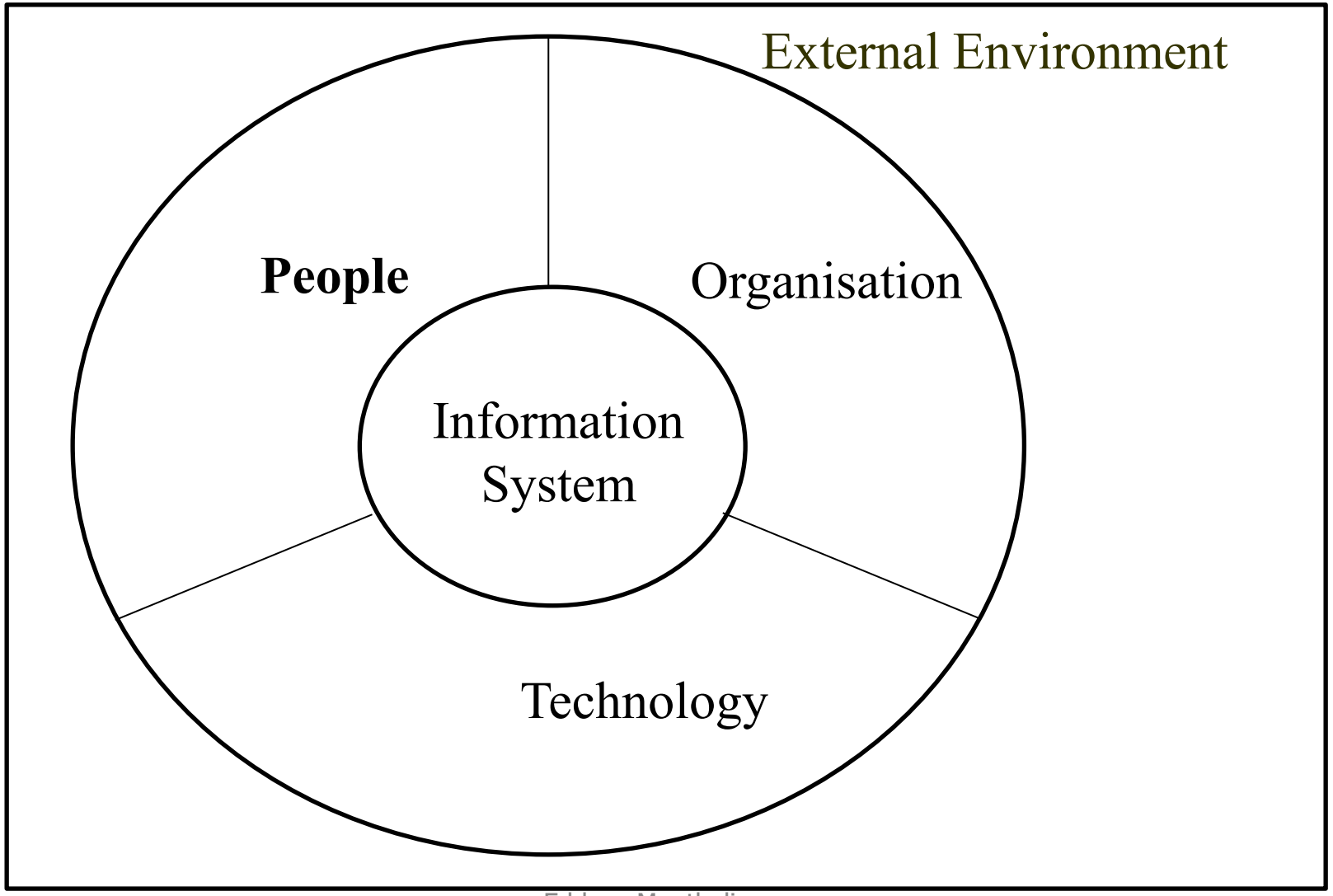
Activities of an IS



Input, Processing, Output.

Input: The activity of gathering and capturing data. Whatever goes into the computer
Processing: Converting or transforming data into useful outputs
Output Useful information, usually in the form of documents and/or reports. Anything that comes out of a computer

Components of an IS



Feedback and Forecasting

- Feedback
  - Output that is used to make changes to input or processing activities
- Forecasting
  - A proactive approach to feedback
  - Use for estimating future sales or inventory needs

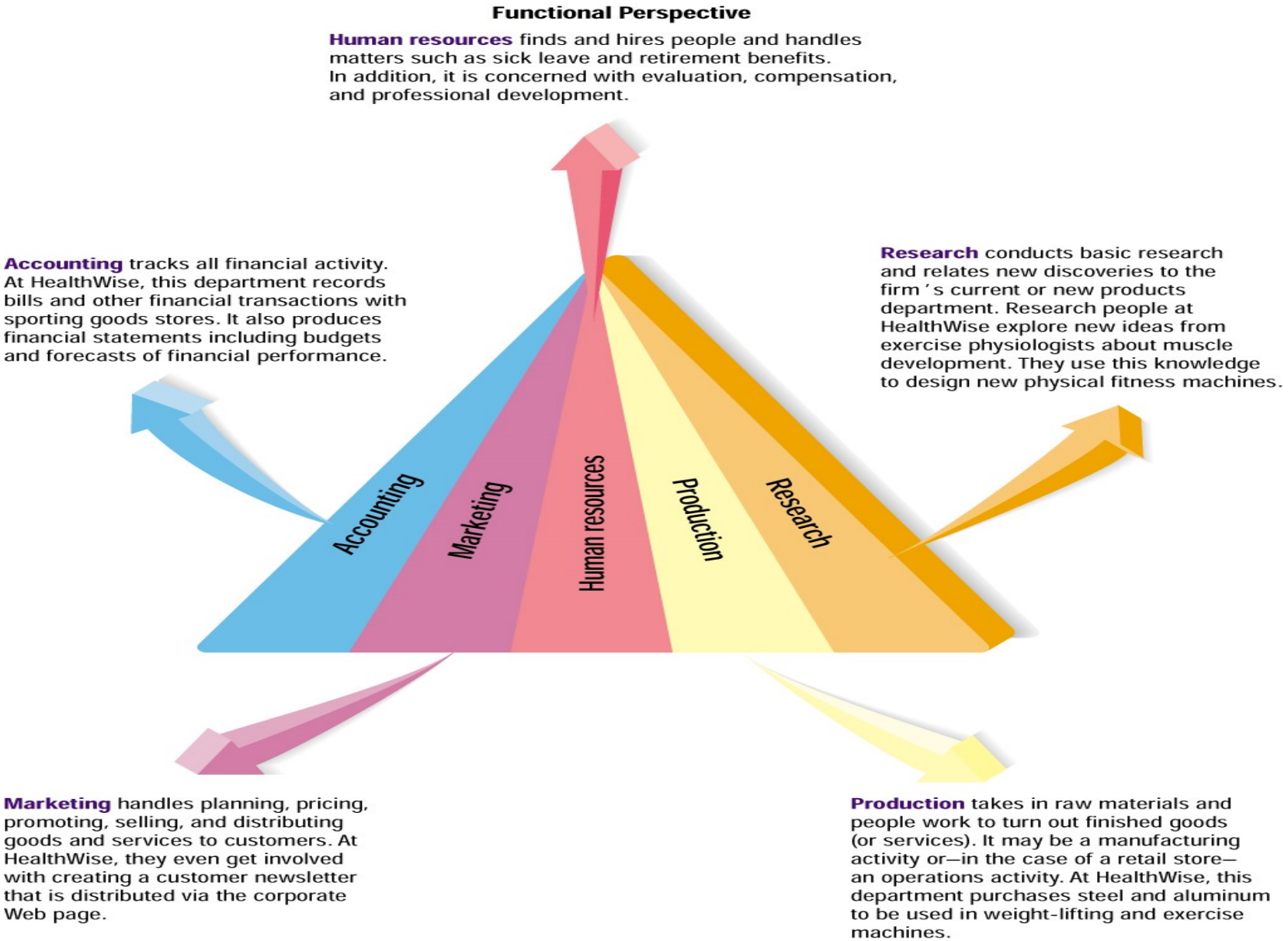
Manual vs. Computerized Systems

- Manual systems still widely used
  - E.g., some investment analysts manual draw charts and trend lines to assist them in making investment decisions
- Computerized systems
  - E.g., the above trends lines can be drawn by computer
- Evolution
  - Many computerized system began as manual systems
  - E.g., directory assistance (“911”)

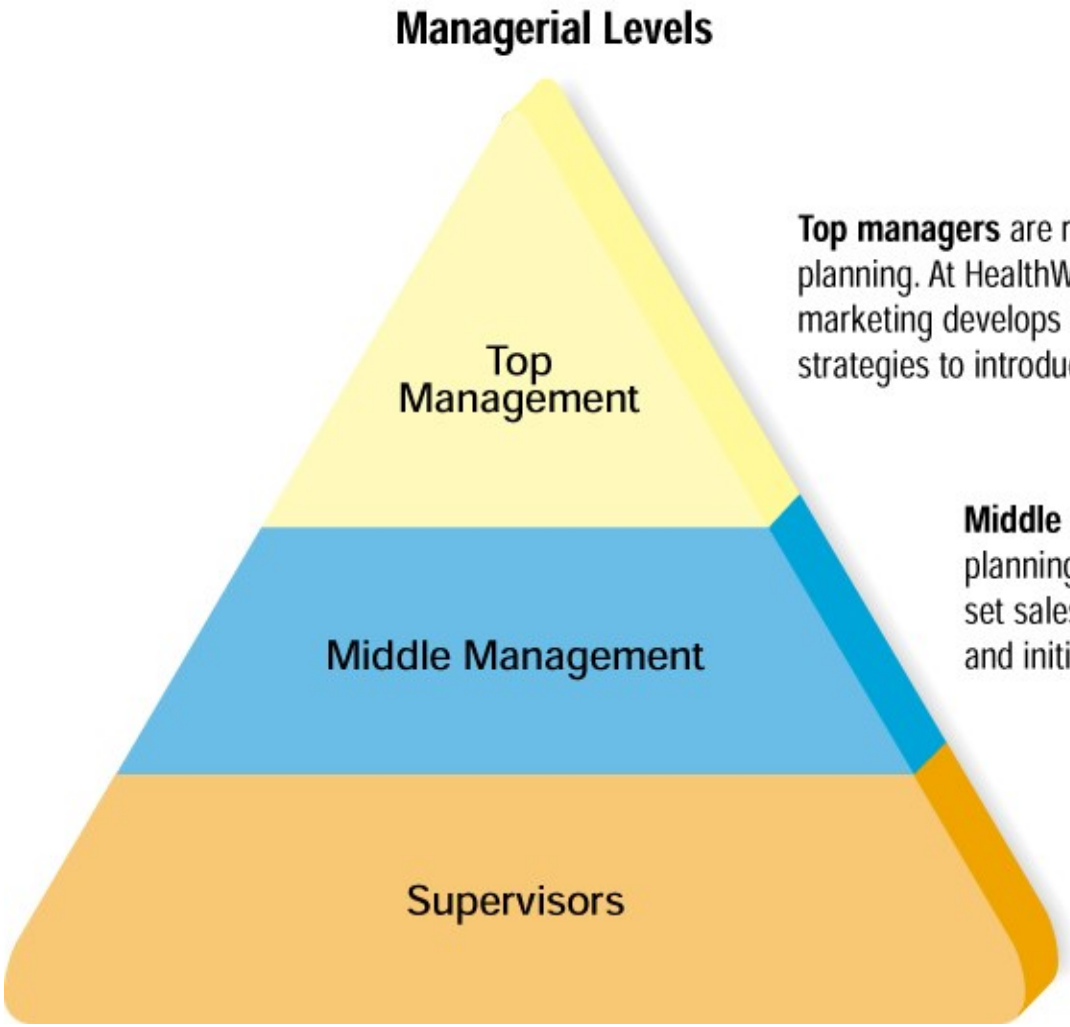
Computer-based Information Systems (CBIS)

- A CBIS is composed (Components) of…
  - Hardware
  - Software
  - Databases
  - Telecommunications
  - People
  - Procedures
- Together they are…
  - Configured to collect, manipulate, store, and process data into information

Organizational Information Flow



Management Levels

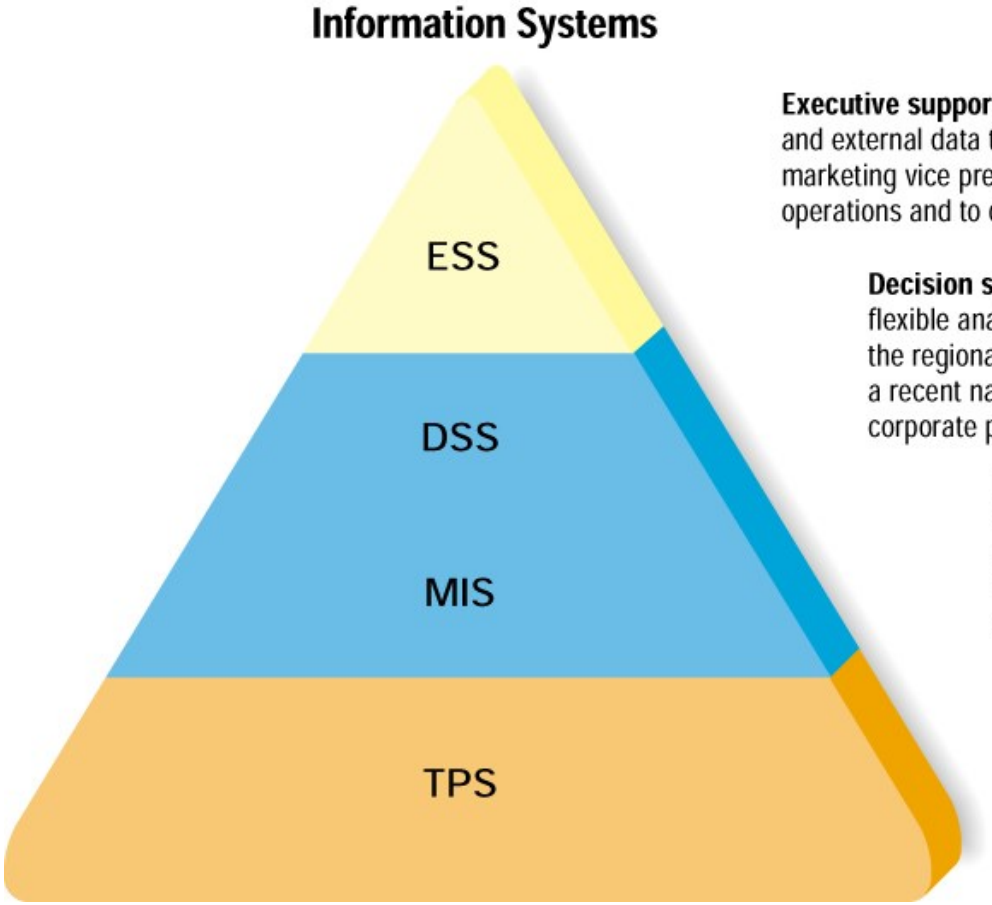


Top managers are responsible for long-range planning. At HealthWise, the vice president of marketing develops long-term marketing strategies to introduce newly developed products.

Middle managers are responsible for tactical planning. At HealthWise, regional sales managers set sales goals, monitor progress to meet goals, and initiate corrective action as needed.

Supervisors are responsible for operational matters. At HealthWise, a production supervisor monitors the inventory for parts and reorders when low.

Computer-Based Information Systems

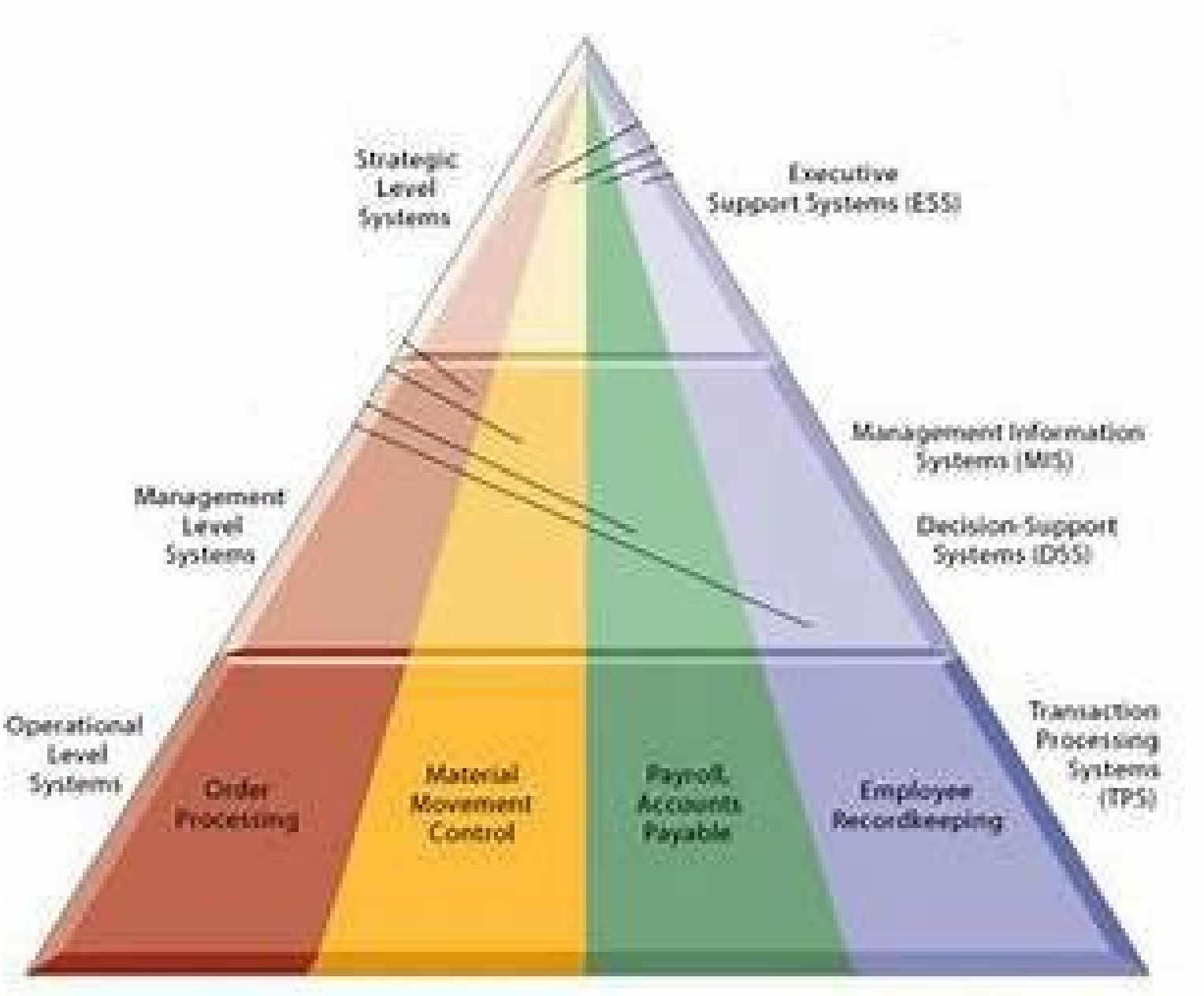


Executive support systems (ESS) use internal data from MIS and TPS and external data to support top-level managers. At HealthWise, the marketing vice president uses his ESS to view current marketing operations and to develop long-term marketing strategies.

Decision support systems (DSS) use data from TPS and a set of flexible analytical tools to support middle managers. At HealthWise, the regional sales managers use the DSS to evaluate the impact of a recent national promotion campaign on regional sales and corporate profit.

Management information systems (MIS) use data from the TPS to support middle-level managers. At HealthWise, regional sales managers use their MIS reports to monitor weekly divisional sales and to compare actual sales to established sales goals.

Transaction processing systems (TPS) record day-to-day transactions to support middle managers. At HealthWise, production supervisors use their TPS to record and to monitor current inventory levels and production line output.


Strategic Level Systems
Executive Support Systems (ESS)
Management Information Systems (MIS)
Management Level Systems
Decision-Support Systems (DSS)
Operational Level Systems
Order Processing
Material Movement Control
Payroll, Accounts Payable
Employee Recordkeeping
Transaction Processing Systems (TPS)

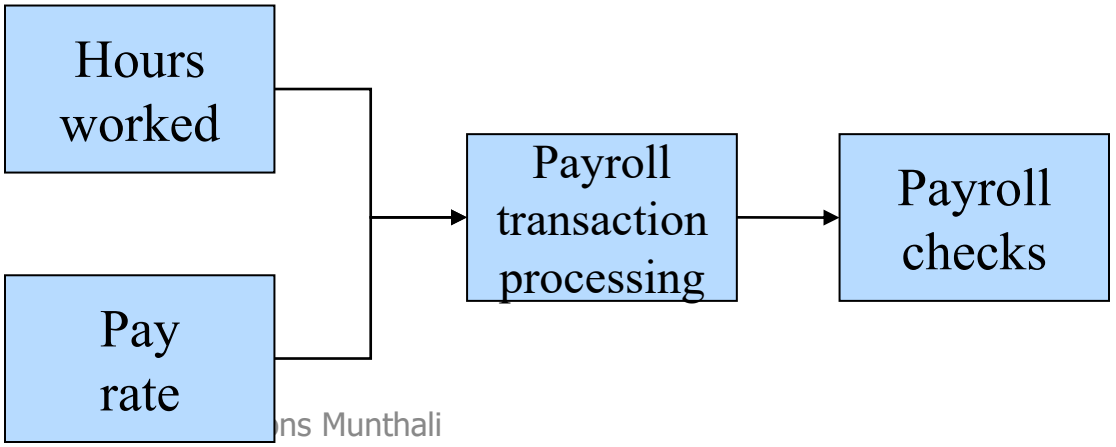
Business Information Systems

- Types
  - Transaction processing systems
  - E-commerce systems
  - Management information systems
  - Decision support systems
  - Expert systems

Transactions Processing Systems

- Transaction
  - Any business-related exchange **(day to day routine activities)**
  - E.g., generating a weekly payroll
- Transaction processing system (TPS)
  - An organized collection of people, procedures, software, databases, and devices used to record completed info for business related exchanges

Payroll Example



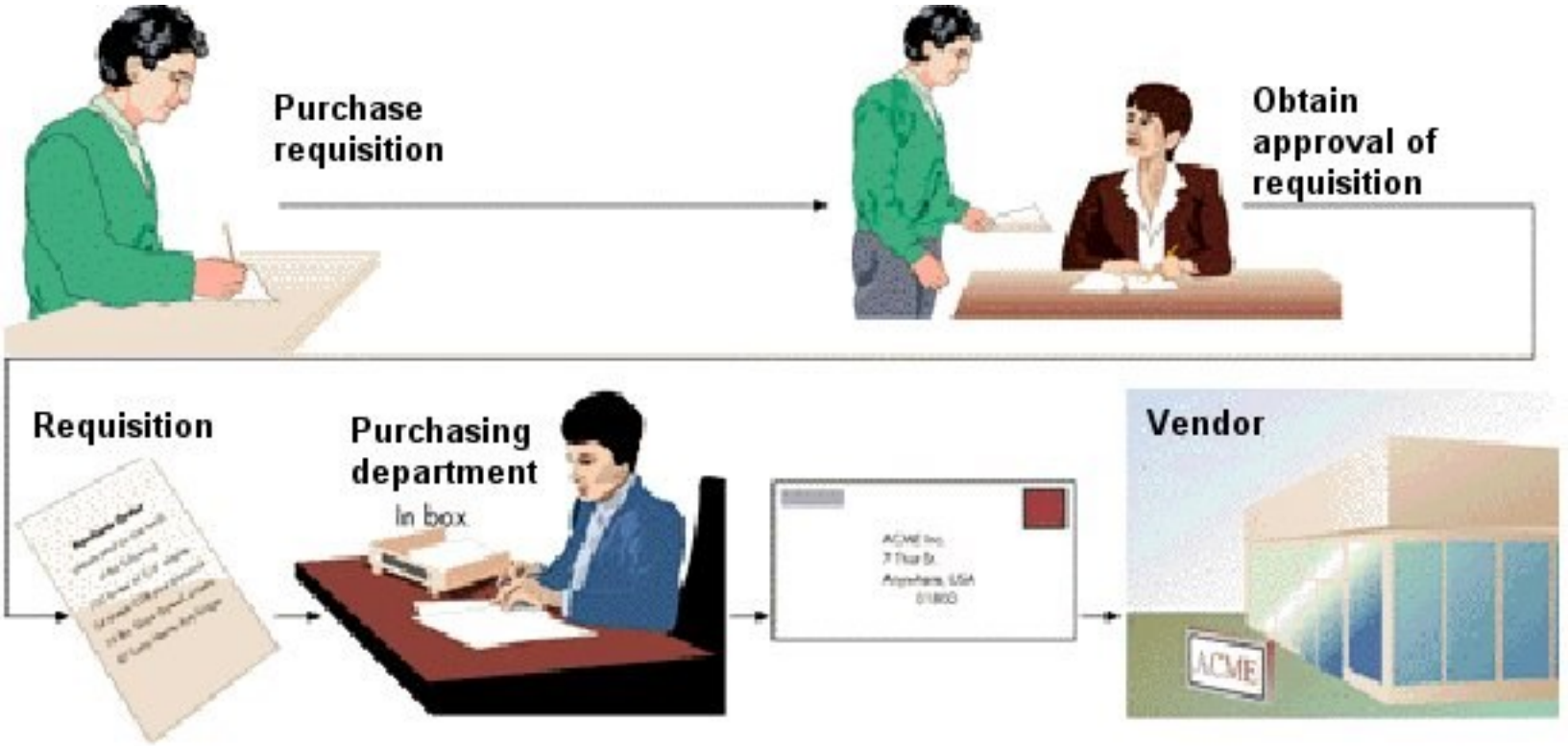
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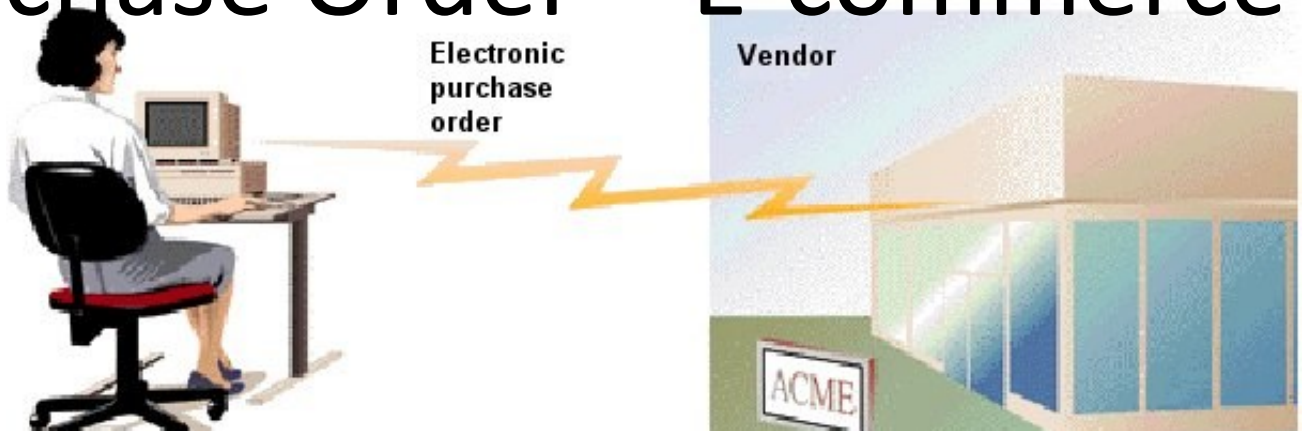
E-Commerce System

- E-commerce
  - Involves any business transaction executed electronically
  - Conducting business on-line
  - For example, between…
    - Companies
    - Companies and consumers
    - Business and the public sector
    - Consumers and the public sector
  - Example for placing a purchase order

Purchase Order - Traditional



Purchase Order – E-commerce



MIS is short for management information system or management information services, and pronounced as separate letters

- **An MIS is…**
  - An organized collection of people, procedures, software, databases, and devices used to record routine information to managers and decision makers

MIS refers to a class of software that **provides managers with tools for organizing and evaluating their department.**

Decision Support Systems

- A DSS is…
  - An organized collection of people, procedures, software, databases, and devices used to support problem-specific decision making
- A DSS helps a manager “do the right thing”

Sensitivity, What-if, and Goal Seeking Analysis

- Sensitivity
  - Assesses impact of change in inputs on outputs
  - Eliminates or reduces variables
  - Can be automatic or trial and error
- What-if
  - Assesses solutions based on changes in variables or assumptions (scenario analysis)
- Goal seeking
  - Backwards approach, starts with goal
  - Determines values of inputs needed to achieve goal
  - Example is break-even point determination

Expert Systems

- An expert system is…
  - A computer application that performs a task that would otherwise be performed by a human expert
  - gives the computer the ability to make suggestions and to act like an expert in a particular field
  - Examples: diagnose human illnesses, make financial forecasts, schedule routes for delivery vehicles

- Expert systems typically include “artificial intelligence” (next slide)

Artificial Intelligence

- AI is…
  - A branch of computer science concerned with making computers behave like humans
  - Term was coined in 1956 by John McCarthy at the Massachusetts Institute of Technology

AI ‘Discussion’(1)

Artificial intelligence includes games playing: programming computers to play games such as chess and checkers

expert systems : programming computers to make decisions in real-life situations (for example, some expert systems help doctors diagnose diseases based on symptoms)

natural language : programming computers to understand natural human languages

neural networks : Systems that simulate intelligence by attempting to reproduce the types of physical connections that occur in animal brains

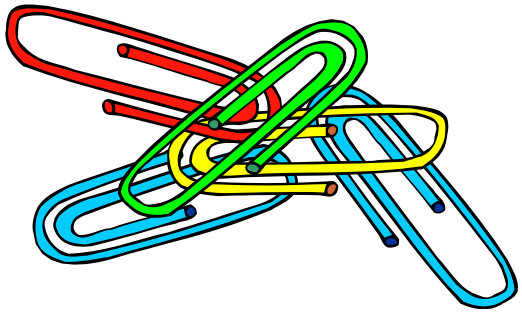
robotics : programming computers to see and hear and react to other sensory stimuli

Problems with Expert Systems (and AI in general)

- Attempted to replace humans in decision making
- However did not take into account
  - How humans actually reason
  - Human information needs (doctors do not want their decision making replaced, but rather want it supported)

Other Information Systems

- Information workers
  - Data workers
    - Secretaries
    - Clerks
  - Knowledge workers
    - Engineers
    - Scientist



- Office automation systems
  - Project managers
  - Videoconferencing systems
- Knowledge work systems
  - Use specialized systems
  - CAD/CAM